Tests for a parallelogram

The properties run backwards: three conditions, any one of which is enough to guarantee that a quadrilateral is a parallelogram.

LESSON 6 | 1 × 40 MIN | GEOMETRY 8, TEMA 3 | STAGE 9 · BEYOND

LESSON CLOCK

4' | 14' | 7' | 13' | 2'

Warm-up	4 min
Explanation	14 min
Interactive	7 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- State the three tests for a parallelogram.
- Tell a property (given a parallelogram, deduce …) from a test (given …, deduce a parallelogram).
- Prove a quadrilateral is a parallelogram using the appropriate test.
- Recognise a condition that is *not* sufficient.

TEXTBOOK

National	Тема 3, pp. 11–13
Cambridge	Extension beyond Stage 9
Workbook	–

01 Explanation

Property or test?

Last lesson every statement began “*in a parallelogram* …”. This lesson every statement ends “… *therefore it is a parallelogram*”. Same facts, opposite direction — and getting the direction right is most of the marks in a proof.

PROPERTY (TEMA 2)	TEST (TEMA 3)
A parallelogram has opposite sides equal.	Opposite sides equal $\Longrightarrow$ parallelogram.
A parallelogram has opposite angles equal.	Opposite angles equal $\Longrightarrow$ parallelogram.
Its diagonals bisect each other.	Diagonals bisecting each other $\Longrightarrow$ parallelogram.

Test 1 — two opposite sides equal and parallel

TEST 1

If two **opposite** sides of a quadrilateral are equal **and** parallel, it is a parallelogram.

Proof. Suppose $AB = CD$ and $AB \parallel CD$. Draw the diagonal AC. Then $\angle BAC = \angle DCA$ (alternate angles) and AC is common, so $\triangle ABC \cong \triangle CDA$ by SAS. Hence $\angle BCA = \angle DAC$, and those are alternate angles for BC and AD — so $BC \parallel AD$ and the definition is satisfied. ∎

BOTH CONDITIONS, ON THE SAME PAIR

Equal *and* parallel, and it must be the *same* pair of sides. Two sides equal and a different pair parallel gives an isosceles trapezium, not a parallelogram.

Test 2 — both pairs of opposite sides equal

TEST 2

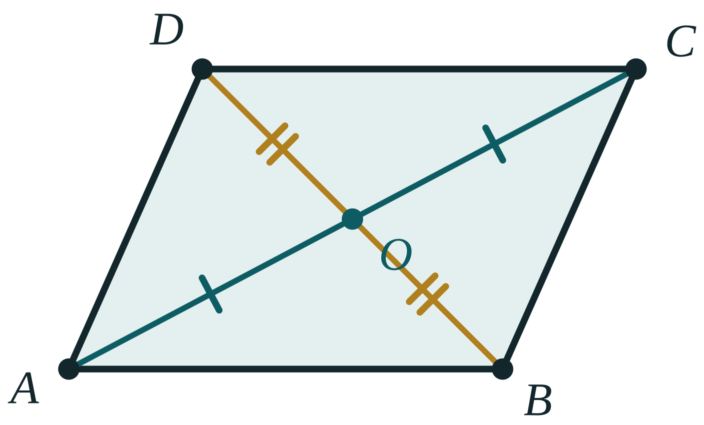
If $AB = CD$ and $BC = AD$, the quadrilateral is a parallelogram.

Proof. The diagonal AC makes $\triangle ABC \cong \triangle CDA$ by SSS. Then both pairs of alternate angles are equal, so both pairs of opposite sides are parallel. ∎

Test 3 — the diagonals bisect each other

TEST 3

If the diagonals of a quadrilateral bisect each other, it is a parallelogram.



AO = OC and BO = OD is enough on its own — the shape has to be a parallelogram.

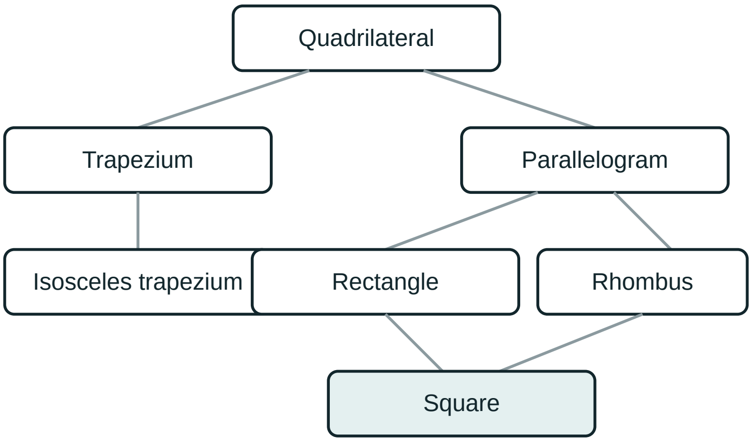
Proof. With $AO = OC$, $BO = OD$ and the vertically opposite angles at O equal, $\triangle AOB \cong \triangle COD$ by SAS. So $AB = CD$, and the equal alternate angles give $AB \parallel CD$. Test 1 finishes the job. ∎

A FOURTH TEST, SOMETIMES USEFUL

If both pairs of opposite **angles** are equal, the quadrilateral is a parallelogram. From $\angle A = \angle C$ and $\angle B = \angle D$ with total $360°$ we get $\angle A + \angle B = 180°$, which makes the sides parallel as co-interior angles.

Conditions that are not enough

- One pair of sides parallel only → a **trapezium**.
- One pair parallel and the *other* pair equal → could be an isosceles trapezium.
- Diagonals equal → could be an isosceles trapezium or a rectangle.
- Two adjacent sides equal → could be a kite.



Where each quadrilateral sits. An arrow means "is a special case of".

02 Worked examples

EXAMPLE 1 ***In quadrilateral ABCD, AB = 7 cm, CD = 7 cm and AB ∥ CD. Is ABCD a parallelogram?***

1. One pair of opposite sides is both equal and parallel.
   That is exactly Test 1.

2. Test 1 applies to the same pair of sides AB and CD ✓
   Check that the pair really is opposite.

3. So $ABCD$ is a parallelogram.

ANSWER

Yes — by Test 1.

EXAMPLE 2 ***The diagonals of ABCD meet at O with AO = OC = 5 and BO = 4, OD = 6. Is it a parallelogram?***

1. $AO = OC$ ✓

   One diagonal is bisected.

2. $BO = 4 \neq 6 = OD$ ✗

   The other is not.

3. Test 3 needs **both** diagonals bisected.

   One is not enough.

ANSWER

No — Test 3 fails.

EXAMPLE 3 ***M is the midpoint of both AC and BD. Prove ABCD is a parallelogram.***

1. $AM = MC$ and $BM = MD$

   M is the midpoint of each diagonal.

2. So the diagonals bisect each other.

   Exactly the hypothesis of Test 3.

3. $ABCD$ is a parallelogram.

ANSWER

Parallelogram, by Test 3.

03 Interactive model

Use the model to test each condition: which ones force the shape, and which leave it free?

Is this enough to force a parallelogram?

> One pair of opposite sides is equal **and** parallel.

1. Draw diagonal AC. Alternate angles give $\angle BAC = \angle DCA$.

   Uses $AB \parallel CD$.

2. $AB = CD$, AC is common, so $\triangle ABC \cong \triangle CDA$ by SAS.

3 Then

$\angle BCA$, so
=
$\angle DAC$

$BC \parallel AD$.

Both pairs are now parallel.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Test (criterion)	Alomat	Признак
Sufficient condition	Yetarli shart	Достаточное условие
Necessary condition	Zarur shart	Необходимое условие
Converse statement	Teskari tasdiq	Обратное утверждение
Counter-example	Qarshi misol	Контрпример
Quadrilateral	To‘rtburchak	Четырёхугольник
Trapezium	Trapetsiya	Трапеция
Kite	Deltoid	Дельтоид

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Which condition forces a quadrilateral to be a parallelogram?

- its diagonals are equal
- its diagonals bisect each other
- two adjacent sides are equal
- one pair of sides is parallel

$AB = CD$ and $AB \parallel CD$ means the quadrilateral is:

- a trapezium
- a parallelogram
- a kite
- not determined

A quadrilateral with one pair of parallel sides only is:

- a parallelogram
- a trapezium
- a rhombus
- a rectangle

Both pairs of opposite angles equal means:

- nothing in particular
- it is a parallelogram
- it is a rectangle
- it is a rhombus

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *$AB = CD$ and $AB \parallel CD$. Is $ABCD$ a parallelogram?*

   ANSWER Yes — Test 1.

2. *$AB = CD$ and $BC = AD$. Is $ABCD$ a parallelogram?*

   ANSWER Yes — Test 2.

3. *The diagonals bisect each other. Is $ABCD$ a parallelogram?*

   ANSWER Yes — Test 3.

4. *Only $AB \parallel CD$ is known. What is $ABCD$?*

   ANSWER A trapezium.

5. *$\angle A = \angle C$ and $\angle B = \angle D$. Is $ABCD$ a parallelogram?*

   ANSWER Yes.

6. *$AC = BD$. Is $ABCD$ a parallelogram?*

   ANSWER Not necessarily.

7. *$AB = BC = CD = DA$. Is $ABCD$ a parallelogram?*

   ANSWER Yes — a rhombus, which is a parallelogram.

Medium · the standard question

7 PROBLEMS

1. *$AO = OC = 5$, $BO = 4$, $OD = 6$. Is ABCD a parallelogram?*

   ANSWER No — only one diagonal is bisected.

2. *$AB = 7$, $CD = 7$, $BC \parallel AD$. Is ABCD a parallelogram?*

   ANSWER Not necessarily — an isosceles trapezium fits.

3. *M is the midpoint of both AC and BD. Prove ABCD is a parallelogram.*

   ANSWER The diagonals bisect each other — Test 3.

4. *$\angle A = 80°$ and $\angle C = 80°$, $\angle B = 100°$. Find $\angle D$ and decide.*

   ANSWER $\angle D = 100°$; opposite angles equal, so it is a parallelogram.

5. *$AB = 6$, $BC = 9$, $CD = 6$, $DA = 9$. Is ABCD a parallelogram?*

   ANSWER Yes — Test 2.

6. *$AB = 6$, $BC = 9$, $CD = 9$, $DA = 6$. Is ABCD a parallelogram?*

   ANSWER No — the equal sides are adjacent; this is a kite.

7. *Which test proves a rhombus is a parallelogram?*

   ANSWER Test 2 — both pairs of opposite sides are equal.

Hard · stretch and reason

7 PROBLEMS

1. *In $\triangle ABC$, M is the midpoint of BC. Point D is on ray AM with $AM = MD$. Prove $ABDC$ is a parallelogram.*

   ANSWER M is the midpoint of both BC and AD, so the diagonals bisect each other — Test 3.

2. *K and L are the midpoints of AB and CD in parallelogram $ABCD$. Prove $AKCL$ is a parallelogram.*

   ANSWER $AK = \frac{1}{2}AB = \frac{1}{2}CD = CL$ and $AK \parallel CL$ — Test 1.

3. *The diagonals of $ABCD$ meet at O and $\triangle AOB \cong \triangle COD$. Must $ABCD$ be a parallelogram?*

   ANSWER Yes if the congruence pairs AO with CO and BO with DO — then the diagonals bisect each other.

4. *Prove that if the opposite angles of a quadrilateral are equal, it is a parallelogram.*

   ANSWER $2\angle A + 2\angle B = 360°$, so $\angle A + \angle B = 180°$ — co-interior angles make the sides parallel.

5. *Give a quadrilateral with $AB = CD$, $AD = BC$ that is not a parallelogram, or explain why none exists.*

   ANSWER None exists — Test 2 proves every such quadrilateral is a parallelogram (for a simple, non-crossed quadrilateral).

6. *E is the midpoint of AB and F of DC in parallelogram $ABCD$. Prove $AEFD$ is a parallelogram.*

   ANSWER $AE = \frac{1}{2}AB = \frac{1}{2}DC = DF$ and $AE \parallel DF$ — Test 1.

7. *In parallelogram $ABCD$, points E and F lie on the diagonal AC with $AE = FC$. Prove $BEDF$ is a parallelogram.*

   ANSWER The diagonals BD and EF share the midpoint O of AC, so they bisect each other — Test 3.

07 Homework

Homework — 5 problems

Geometry 8, Тема 3, pp. 11–13. Name the test you used in each proof.

1. *$AB = 9$, $CD = 9$ and $AB \parallel CD$. Prove $ABCD$ is a parallelogram and name the test.*
2. *$AO = 4$, $OC = 4$, $BO = 7$, $OD = 7$. Prove $ABCD$ is a parallelogram.*
3. *Give an example of a quadrilateral with equal diagonals that is not a parallelogram.*
4. *In $\triangle ABC$, M is the midpoint of AC and BM is extended to D with $BM = MD$. Prove $ABCD$ is a parallelogram.*
5. Write the three tests in your own words, each with a small labelled sketch.